Protecting Images from Diffusion-Based Geolocation

How to prevent cheating at Geoguesser

Mathias Ollu, Elliot Thorel (Ecole Polytechnique)

Motivation

- **Confidentiality:** how to hide an image's location ?
- **Robustness:** can geographic diffusion be attacked ?
- **Attack strategy:** what to attack in a diffusion model ?

Existing works

- Diffusion/flow matching visual geolocation (Dufour et al., 2024)
- Geoprivacy against VLMs (Liu et al., 2025)
- Adv. attacks vs. Diffusion models (Salman et al., 2023).

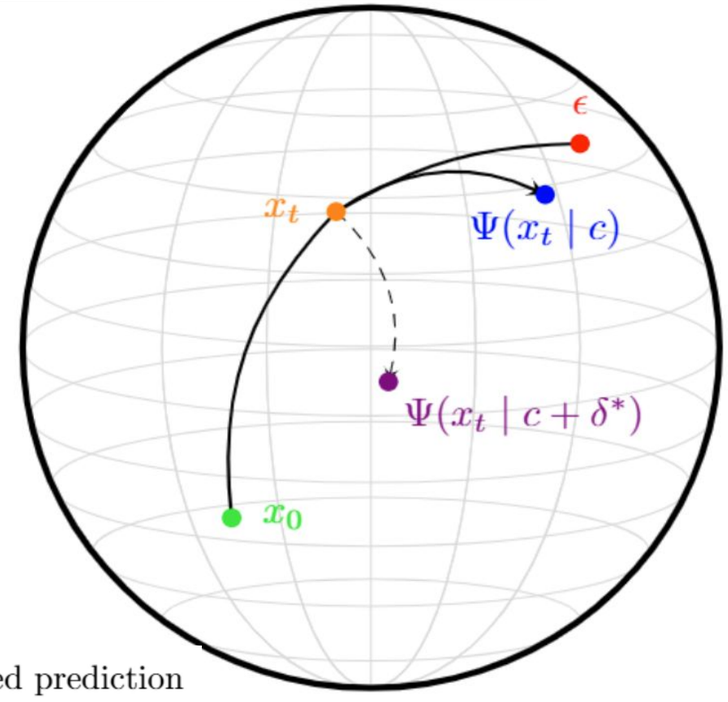
  Achieves stealth: $\frac{16}{255}$

Dufour et al. training objective

$$\psi^* = \underset{\psi}{argmin}\,\mathbb{E}_{x_0,c,\epsilon,t}[||\psi(x_t|c) - \epsilon||^2]$$

Intuition:
Attacking diffusion is **hard**

→ cause **minor disruptions** that accumulate through diffusion process



- x_0: true location
- ϵ: sampled noise
- x_t: noisy location
- $\psi(x_t \mid c)$: prediction
- $\psi(x_t \mid c + \delta^*)$: optimal perturbed prediction

Our methodology

Likelihood test

Flat likelihood approach

- $\log p(x_i \mid c_i) > \theta$

Use of localizability

- $\log p(x_i \mid c_i) - \int \log p(x \mid c_i)\, p_{\text{model}}(x \mid c_i)\, dx > \delta$

Test datasets were generated from the YFCC dataset. Half of the images were assigned incorrect locations, produced using one of two different methods:

1) Assigning to an image the location of another, randomly chosen image.
2) Selecting a random location within a circular region of radius R around the true location.

Attack setup : $(x_0, c) \sim \Omega, \epsilon \sim \mathcal{N}(0, I), t \sim \mathcal{U}[0,1]$

Encoder attack (Salman et al., 2023)

$$\delta_{encoder} = \underset{||\delta||_\infty \leq b}{argmin}||\Sigma(c+\delta) - z_{targ}||^2$$

$$\delta_{encoder} = \underset{||\delta||_\infty \leq b}{argmax}||\Sigma(c+\delta) - \Sigma(c)||^2$$

Diffusion Trajectory Deviation (DTD - ours)

$$\delta_{DTD} = \underset{||\delta||_\infty \leq b}{argmin}\mathbb{E}_{x_0,c,\epsilon,t}[|\psi(x_t|c_{src} + \delta) \cdot \epsilon|]$$

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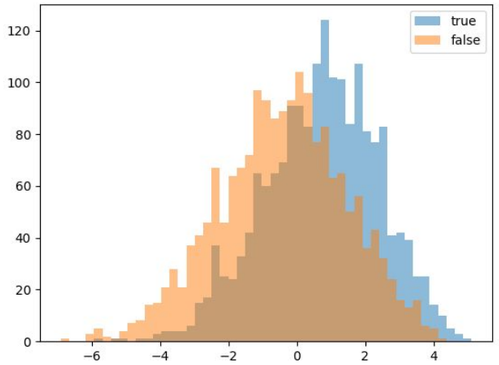
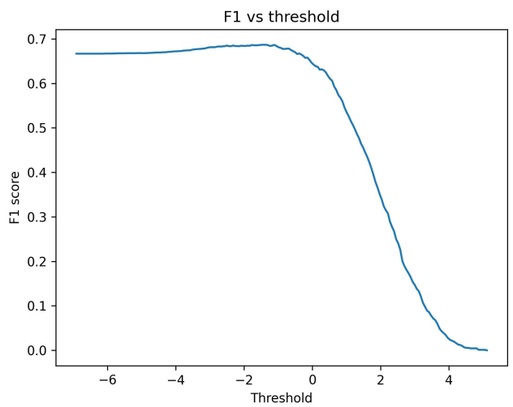
Experimental setting

- Datasets: **YFCC** and **OSV-5m** (test)
- Encoders: **DinoV2** and **ViT finetuned on StreetClip**
- Model archi: 6 blocs of **MLP+GELU,** conditioning via **AdaLN**

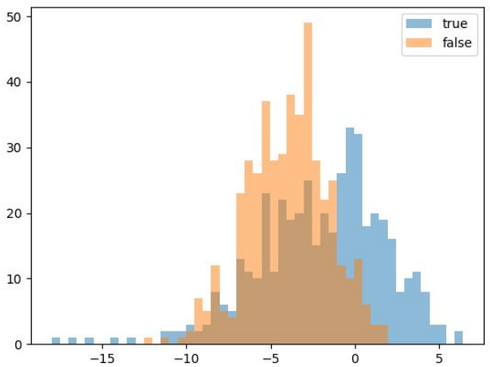
Detecting image loc. manipulation

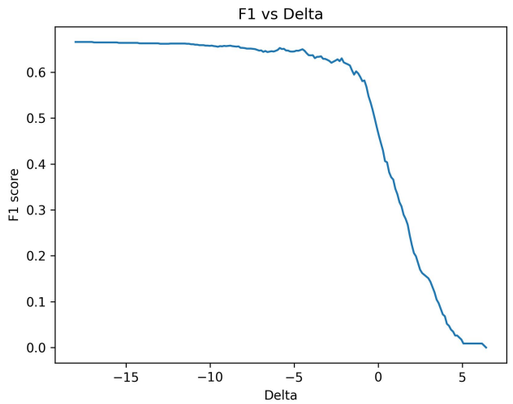
Flat likelihood approach



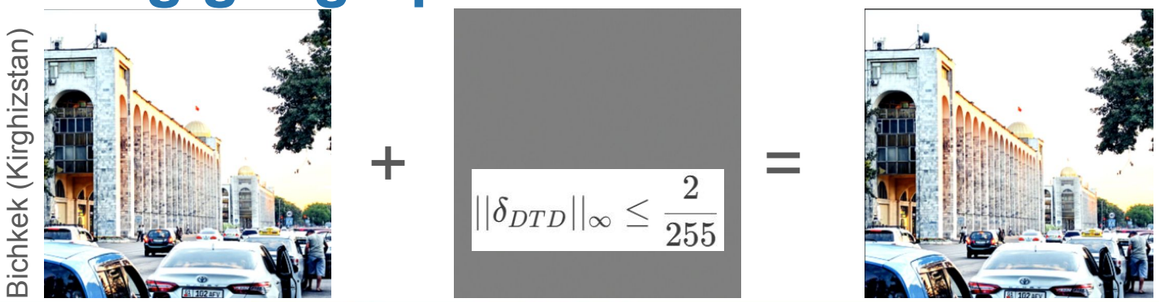


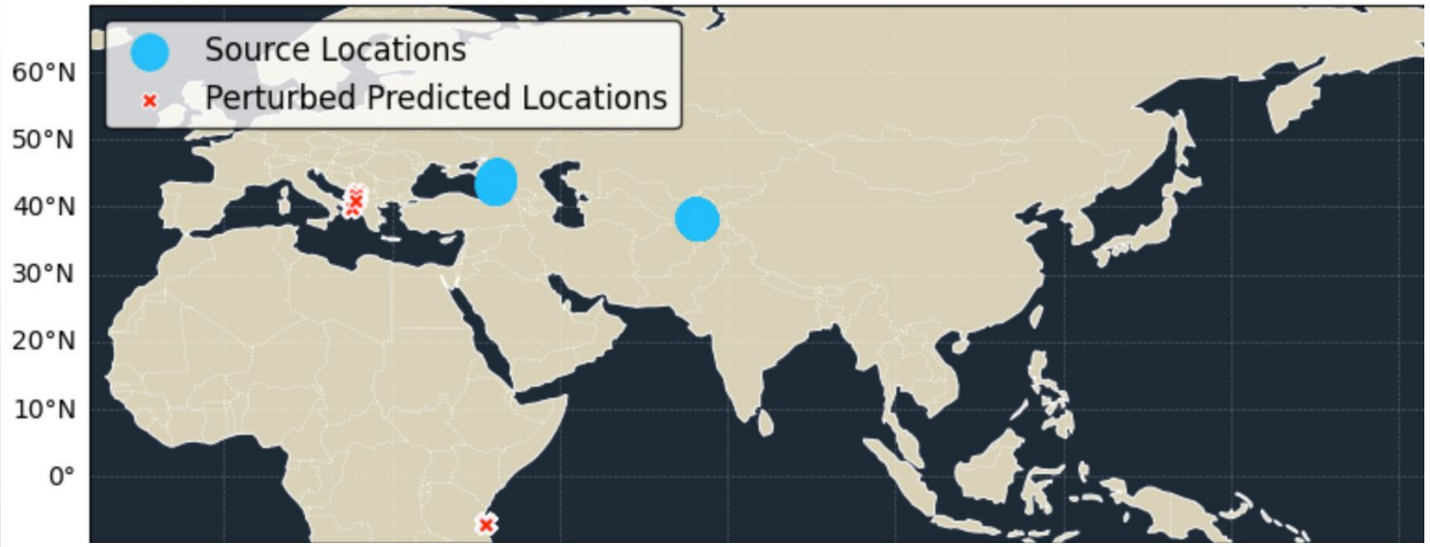
Use of localizability



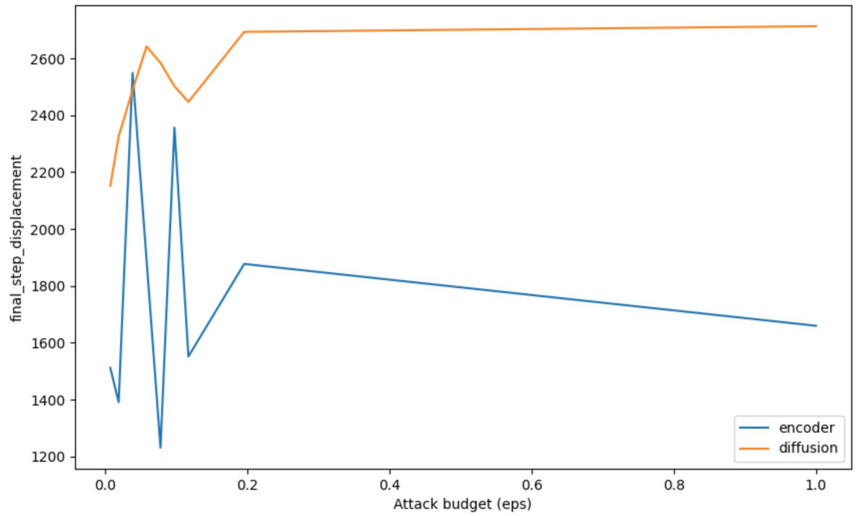


Hacking geographic diffusion





Encoder vs DTD attacks
(haversine dist, km)



Implications

- **Geolocation diffusion is easy to hack: privacy, deepfakes…**
- **Targeting diffusion process is cheap + effective (+ transferable)**
- **Lessons for image/text/robotic manipulation diffusion?**